Practical No 05

a. Develop an android application demonstrating Threads and Handlers for Background processing.

ProjectName: ThreadHandlerDemo

FileName: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">
    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Background Task"/>
    <TextView
        android:id="@+id/txtResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="22sp"
        android:paddingTop="20dp"/></LinearLayout>
```

FileName: MainActivity.java

```
package com.example.threadhandlerdemo;


import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.TextView;


import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    Button btnStart;
    TextView txtResult;


    Handler handler = new Handler(Looper.getMainLooper());


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        btnStart = findViewById(R.id.btnStart);
        txtResult = findViewById(R.id.txtResult);


        btnStart.setOnClickListener(v -> {


            // Background Thread
            Thread thread = new Thread(() -> {
                try {
                    // Simulate Background Work
                    Thread.sleep(3000);


                } catch (Exception e) {
                    e.printStackTrace();
                }


                // Update UI using Handler
                handler.post(() -> {
                    txtResult.setText("Background Task
Completed");
                });
            });


            thread.start();
        });
    }
}
```

b. Develop an android application implementing Media APIs and Telephone APIs.

ProjectName: MediaTelePhoneDemo

FileName: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">
    <Button
        android:id="@+id/btnPlay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Music"/>
    <Button
        android:id="@+id/btnDial"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Dialer"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

FileName: MainActivity.java

```
package com.example.mediatelephonedemo;


import android.content.Intent;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;


import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    Button btnPlay, btnDial;
    MediaPlayer mediaPlayer;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        btnPlay = findViewById(R.id.btnPlay);
        btnDial = findViewById(R.id.btnDial);


        // Media API
        mediaPlayer = MediaPlayer.create(this, R.raw.music);
```

```
        // Play music
        btnPlay.setOnClickListener(v -> {
            mediaPlayer.start();
        });


        // Open phone dialer
        btnDial.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:9876543210"));
            startActivity(intent);
        });
    }


    @Override
    protected void onDestroy() {
        super.onDestroy();

        if (mediaPlayer != null) {
            mediaPlayer.release();
            mediaPlayer = null;
        }
    }
}
```

C. Develop an Android application implementing RunTime Permission and security handling in AndroidManifest.xml.

Project Name: PermissionDemo

FileName: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <Button
        android:id="@+id/btnPermission"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Request Camera Permission"/>
</LinearLayout>
```

FileName: MainActivity.java

```
package com.example.permissiondemo;


import android.content.pm.PackageManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    Button btnPermission;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        btnPermission = findViewById(R.id.btnPermission);


        btnPermission.setOnClickListener(v -> {
            checkPermission();
        });
    }


    private void checkPermission() {
```

```
        // Check if Camera permission is already granted
        if (ContextCompat.checkSelfPermission(
                this,
                android.Manifest.permission.CAMERA
        ) == PackageManager.PERMISSION_GRANTED) {


            Toast.makeText(
                    this,
                    "Permission Already Granted",
                    Toast.LENGTH_SHORT
            ).show();


        } else {


            // Request Camera permission
            ActivityCompat.requestPermissions(
                    this,
                    new String[]{
                            android.Manifest.permission.CAMERA
                    },
                    1
            );
        }
    }


    // Permission Result
    @Override
    public void onRequestPermissionsResult(
            int requestCode,
            String[] permissions,
            int[] grantResults
    ) {
        super.onRequestPermissionsResult(
                requestCode,
                permissions,
                grantResults
        );


        if (requestCode == 1) {


            if (grantResults.length > 0
                    && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {


                Toast.makeText(
                        this,
                        "Permission Granted",
                        Toast.LENGTH_SHORT
                ).show();


            } else {
```

```
                Toast.makeText(
                        this,
                        "Permission Denied",
                        Toast.LENGTH_SHORT
                ).show();
            }
        }
    }
}
```

FileName: Android_Manifest.xml

```
package com.example.permissiondemo;


import android.content.pm.PackageManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    Button btnPermission;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        btnPermission = findViewById(R.id.btnPermission);


        btnPermission.setOnClickListener(v -> {
            checkPermission();
        });
    }


    private void checkPermission() {


        // Check if Camera permission is already granted
        if (ContextCompat.checkSelfPermission(
                this,
                android.Manifest.permission.CAMERA
        ) == PackageManager.PERMISSION_GRANTED) {


            Toast.makeText(
                    this,
                    "Permission Already Granted",
```

```
                    Toast.LENGTH_SHORT
            ).show();


        } else {


            // Request Camera permission
            ActivityCompat.requestPermissions(
                    this,
                    new String[]{
                            android.Manifest.permission.CAMERA
                    },
                    1
            );
        }
    }


    // Permission Result
    @Override
    public void onRequestPermissionsResult(
            int requestCode,
            String[] permissions,
            int[] grantResults
    ) {
        super.onRequestPermissionsResult(
                requestCode,
                permissions,
                grantResults
        );


        if (requestCode == 1) {

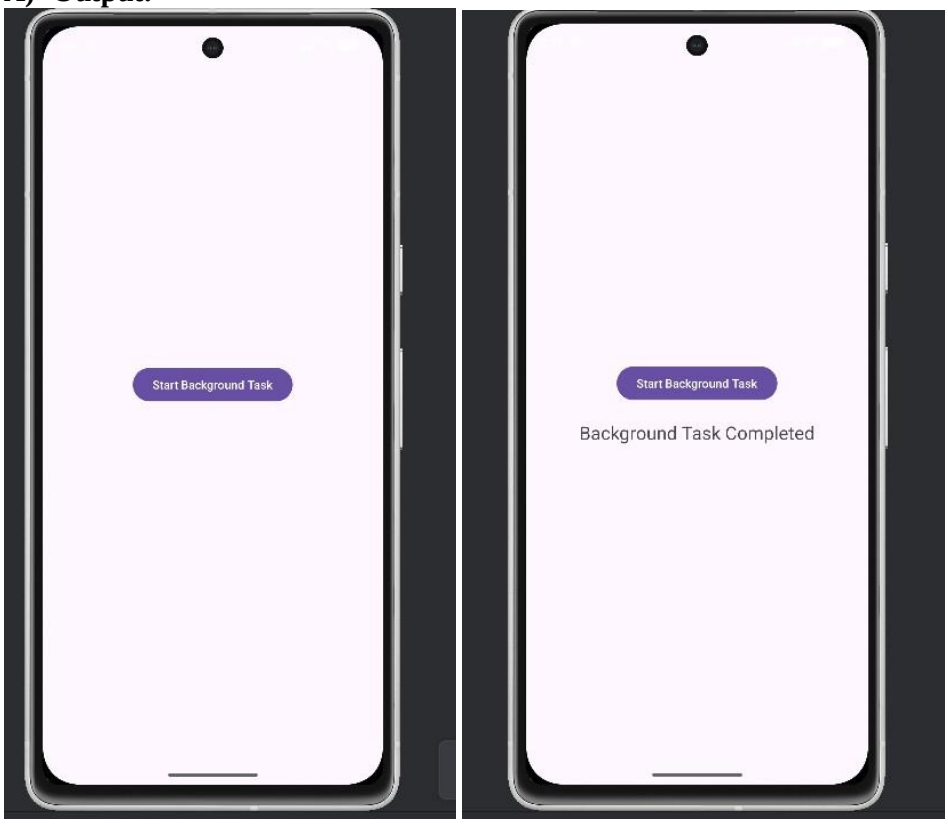

            if (grantResults.length > 0
                    && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {


                Toast.makeText(
                        this,
                        "Permission Granted",
                        Toast.LENGTH_SHORT
                ).show();

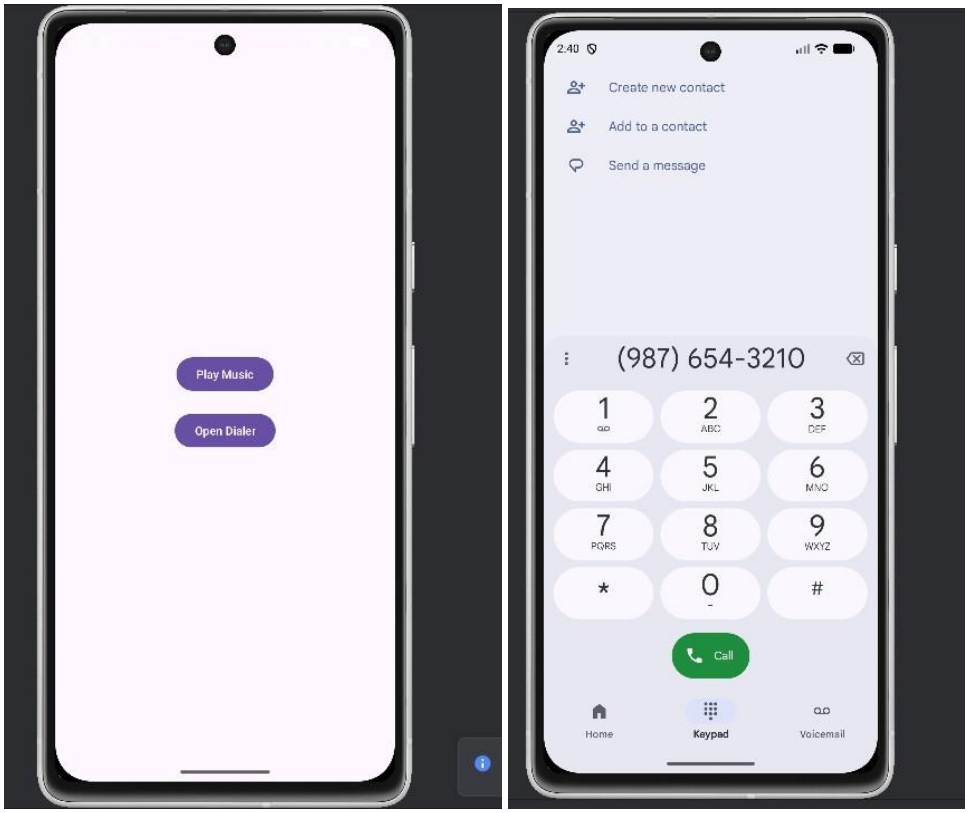

            } else {


                Toast.makeText(
                        this,
                        "Permission Denied",
                        Toast.LENGTH_SHORT
                ).show();
            }
        }
    }
}
```

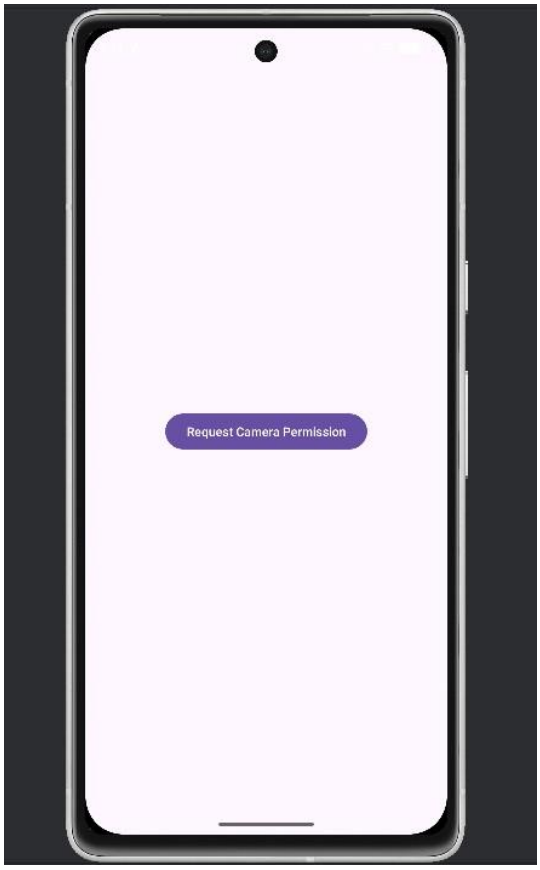
A) Output:

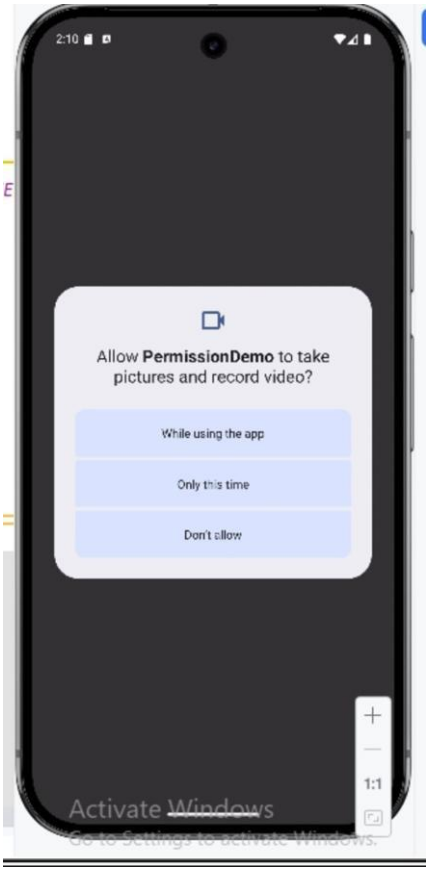

Start Background Task
Start Background Task
Background Task Completed

B) Output :


Play Music
Open Dialer
2:40
Create new contact
Add to a contact
Send a message
(987) 654-3210
1
2 ABC
3 DEF
4 GHI
5 JKL
6 MNO
7 PQRS
8 TUV
9 WXYZ
*
0 +
#
Call
Home
Keypad
Voicemail

C) Output :


Request Camera Permission


2:10
Allow PermissionDemo to take pictures and record video?
While using the app
Only this time
Don't allow
1:1
Activate Windows